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Severe remagnetization revealed from Triassic platform carbonates near Guiyang, Southwest China

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Abstract

An intensive paleomagnetic investigation has been conducted on the Middle and Lower Triassic platform carbonates near Guiyang, Southwest China in the hope of refining a Middle Triassic paleomagnetic pole previously reported from the region. Instead, severe late Cretaceous to Recent remagnetizations have been revealed from all the three rock formations studied. In all cases, the in situ characteristic remanent magnetization (ChRM) directions resemble or fall close to the local present-day Earth's magnetic field direction and fail the fold test, despite the presence of dual polarity in two formations. The major magnetic carriers of the remagnetizations are hematite and magnetite. The cause(s) for the remagnetization events cannot be ascertained at present, but weathering in recent time and migrating fluid in the past might be responsible. It is suggested that caution should be exercised when investigating platform carbonate rocks which are prone to late remagnetization.

Keywords: Triassic; paleomagnetism; carbonate platforms; Guizhou China

1. Introduction

Chan et al. [1] reported a paleomagnetic pole derived from the carbonate rocks of the Middle Triassic Huaxi (Huachi) Formation near Guiyang, the capital of Guizhou province in Southwest China, which was later used by some authors [2] to construct the apparent polar wander path (APWP) for the Yangtze Block (YB). However, this pole was based upon only eight samples and the direction after thermal demagnetization at 200°C was chosen to represent the stable direction for individual samples. This does not meet modern reliability criteria for paleomagnetic studies (e.g. [3]) even though two polarities were present and the fold test appeared to be positive in these eight samples. Realizing that reliable Middle Triassic paleomagnetic data are lacking for the YB and hoping that we could refine the pole Chan and his collaborators reported, we recently conducted an intensive paleomagnetic investigation on two Middle Triassic formations and one Lower Triassic formation from the same region; instead, the results presented in this paper reveal pervasive remagnetization from these platform carbonates.

2. Geological setting

Triassic carbonate rocks cover a large portion of Guizhou province. These rocks are situated in a

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subtropical zone with plentiful precipitation and warm annual temperatures. Thus, intensive karstification coupled with episodic uplift since the Tertiary has shaped these rocks into spectacular landscapes [4]. The Triassic sequence in the vicinity of Guiyang is more than 2800 m in thickness; as in most of the Upper Yangtze region, it is characterized by neritic carbonates in the Lower and Middle Series and is dominated by paralic clastics in the Upper Series [5]. The Lower Triassic in Guiyang is divided into the Daye and Anshun Formations. The Daye Formation is 111–582 m thick and is composed of gray/dark gray thin-bedded limestone and argillaceous limestone interbedded with shale. The rocks are interpreted as restricted platform deposits based upon their thin-bedded nature and the characteristic small size bivalves they contain [6]. The Anshun Formation conformably overlies the Daye Formation and varies between 280 and 712 m in thickness. It is composed of light gray, medium bedded limestone in the lower part, thick bedded dolostone and oolitic dolostone in the middle part, and yellowish/brick-red, thin bedded argillaceous dolostone, brecciated dolostone and salt-solution breccia in the upper part. These rocks were presumably deposited in a shelf lagoon environment. The Middle Triassic Series includes three formations, in ascending order, the Huaxi, Longtou and Gaicha Formations. The Huaxi Formation is 435–900 m thick and consists chiefly of gray, medium bedded dolostone and limestone and their argillaceous varieties. It is conformable with, and separated by a 3–5 m thick, vitric tuffite from the Anshun Formation. The Longtou Formation is made of barrier reef deposits with light gray, thick bedded micritic dolostone in the lower part and alternating laminated dolostone and dolomitic limestone with sparry algal limestone, oncolitic limestone and coagulated limestone in the middle and upper part. It varies in thickness from 174 to 1254 m and yields abundant bivalves, brachiopods, gastropods, crinoids and algae. The Gaicha Formation is 50–95 m thick and is composed mainly of gray, thick bedded micritic dolostone and limestone with a bioclastic limestone bed at the top. The Upper Triassic is represented by the Sanqiao and Erqiao Formations. The Sanqiao Formation disconformably, locally conformably, overlies the Gaicha Formation, and is composed of two cycles from calcareous sandstone, siltstone and carbonaceous shale to limestone, argillaceous limestone and bioclastic limestone with the thickness being 0–90 m. The Erqiao Formation disconformably overlies the Sanqiao Formation, and consists of black shale interbedded with sandstone in the lower part and quartz sandstone in the upper part. It has a similar thickness (30–99 m) to the Sanqiao Formation and yields plant fossils and thin coal seams. These two formations are regarded as littoral swamp and lacustrine swamp deposits, respectively. The ages of the rock formations described above all have very good fossil control.

The paleomagnetic samples of this study are from the Anshun, Huaxi and Longtou Formations. In order to perform a fold test to constrain the age of magnetization, the samples were collected from different flanks of the folds (Fig. 1). Folding of the Triassic rocks in the region is believed to have occurred during early the Cretaceous [6].

3. Paleomagnetic sampling and results

3.1. The Longtou Formation

The Middle Triassic Longtou Formation was sampled in 1994. Samples were collected from two limbs of the Guiyang syncline, a S–N-trending, steeply dipping symmetric fold with the Jurassic rocks in the core and Triassic rocks cropping out on the limbs. Eleven sites were drilled on the west limb near a town called Huaxi (Locality 1 in Fig. 1) using a gasoline-powered rock drill, each site consisting of six samples oriented in situ using a magnetic compass. The rocks at these sites dip to 78–89° east at 65–73°. Eleven more sites were drilled on the east limb near a village called Nandaguan (Locality 2 in Fig. 1) about 1.5 km south of the South Train Station of Guiyang. Ten samples from one site were drilled from clasts of an intrasparitic limestone bed. The rocks on the east limb dip to 263–293° west at 54–85°. Altogether, 135 samples were collected, mostly from crystalline limestones.

The natural remanent magnetization (NRM) of the samples was first measured in a 2G cryogenic magnetometer in the paleomagnetic laboratory of the University of Florida. It was immediately found that the majority of these samples were very weakly

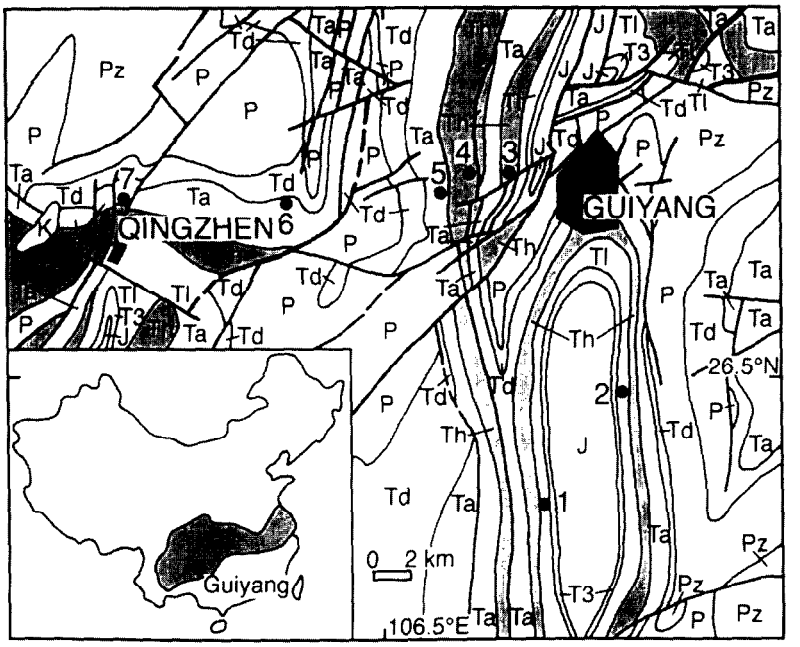


Fig. 1. Sketch geological map of the Guiyang area. Numbered dots denote the sampled localities in this study. Heavy lines are faults. Strata: *Pz* = pre-Permian; *P* = Permian; *Td* = Lower Triassic Daye Fm; *Ta* = Lower Triassic Anshun Fm; *Th* = Middle Triassic Huaxi Fm (shaded); *Tl* = Middle Triassic Longtou and Gaicha Formations; *T3* = Upper Triassic Sanqiao and Erqiao Formations; *J* = Jurassic; *K* = Cretaceous. The shaded area in the inset is the Yangtze Block.

magnetic, with the initial NRM intensities of 82 samples being less than 0.1 mA/m, and that the in situ NRM directions of these samples were tightly clustered around the present geocentric axial dipole field direction at Guiyang (D/I = 0°/45°) with a mean of D/I = 1.4°/47.0° (Fig. 3 upper). The NRM

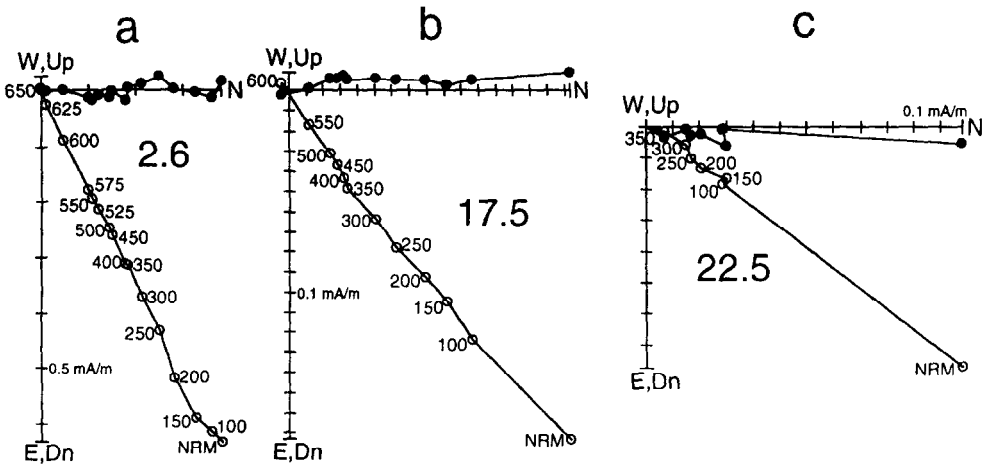


Fig. 2. Representative orthogonal plots of thermal demagnetization (in in situ coordinates) for the Longtou Fm. ● = plotted on horizontal plane; ○ = plotted on vertical plane. Treatment levels are in degrees Celsius.

directions split into two distinct groups controlled by the fold limbs after structural correction with $\kappa_s/\kappa_g = 0.2$ (Fig. 3 upper), a negative fold test indicating that the NRMs of these samples are indeed severely overprinted by the present-day Earth's magnetic field. The 53 samples with initial NRM intensities higher than 0.1 mA/m were then chosen to be further thermally demagnetized. Of these, 47 yielded ChRM directions [7]; all of them exhibited univectorial decay behavior with progressively higher temperatures and all of them are of normal polarity (Fig. 2). The ChRM directions in most of these samples persisted up to above 600°C, suggesting that hematite is a contributor to the NRM (Fig. 2a,b); there was a big drop in NRM intensity after being demagnetized at 100°C in a few samples, suggesting the existence of goethite in these samples (Fig. 2c). It is no surprise that the stereographic projection of the ChRM directions of these samples, calculated using the principal component analysis technique [8], is simply a duplicate of the NRM directions of the whole collection (Fig. 3 lower), indicating that the ChRM directions of these samples represent a recent overprint of the Earth's magnetic field. The mean in situ ChRM direction based on these 47 samples is D/I = 0.7°/51.8° with $\alpha_{95} = 1.9°$, corresponding to a paleomagnetic pole at 84.0°N/112.2°E.

3.2. The Huaxi Formation

The failure to recover the primary magnetization from the Longtou Formation prompted us to consider that it might be wise to resample the Huaxi Formation. The Huaxi Formation was thus resampled in 1995 under the guidance of Professor Xue Yi Wu, who helped Chan collect the samples from the same formation in 1980. Sampling was carried out basically along the same section from Sanqiao to Yey-

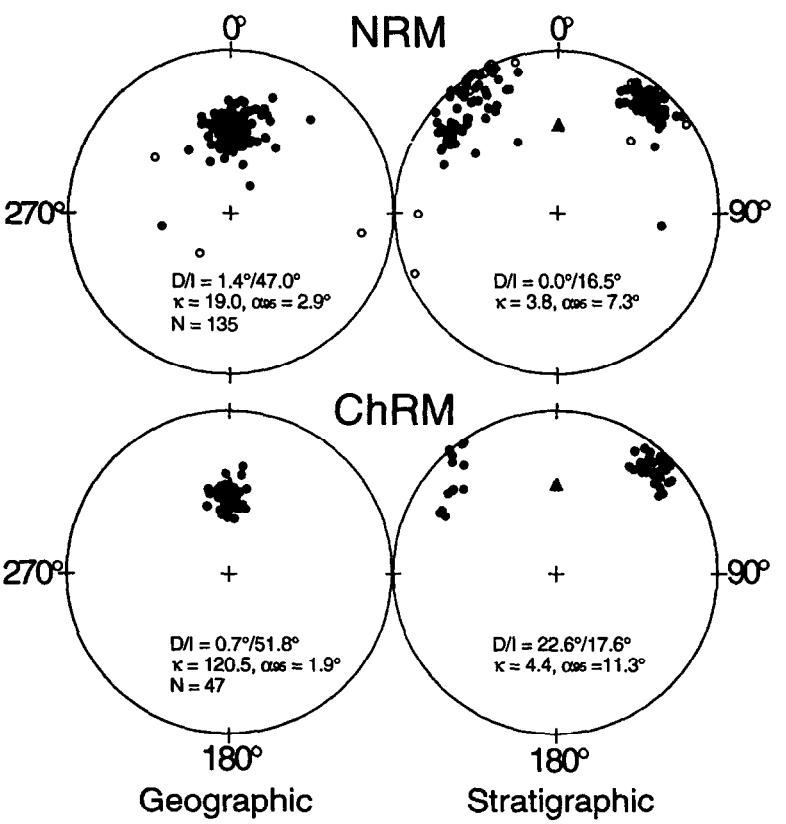


Fig. 3. Equal-area projections of NRM and ChRM directions of the samples from the Longtou Fm. ● = plotted onto the lower hemisphere; ○ = plotted onto the upper hemisphere; ▲ = the present dipole field direction in Guiyang.

atang, west of downtown Guiyang, Chan et al. [1] sampled. The structure is a roughly N–S-trending anticline with a strike fault cutting the west limb and causing the beds to be locally upright to overturned. At the east limb of the anticline, eleven sites were drilled from the gray, medium bedded dolostones, which dip toward 98–118° east at 45–60° (Locality 3 in Fig. 1). At the west limb, nine sites were collected with 6 sites from dolostones and 3 sites from limestones (Locality 4 in Fig. 1). The bedding attitudes at the six dolostone sites vary from dipping to 267–269° west at 61–67° (normal) to dipping to 110–114° east at 74–89° (overturned), and the rock beds at the three limestone sites dip to 275–290° west at 47–54°. A total of 128 samples were collected.

Most of the samples from the Huaxi Formation, especially those from the west limb of the anticline, were more strongly magnetized than those from the Longtou Formation with the initial NRM intensities of 71 samples being higher than 0.1 mA/m. All the samples from this formation were subjected to progressive thermal demagnetization. Most samples yielded ChRM directions, and again nearly all of them univectorially decayed to the origin of coordinates with increasing temperatures (Fig. 4). The samples from the east limb of the anticline had higher unblocking temperatures above 625°C, suggesting that hematite is probably the major magnetic carrier (Fig. 4a–c): the samples from most of the west limb sites, on the other hand, exhibited much lower unblocking temperature spectra, usually below 450°C, indicating that magnetite may be a major contributor to the ChRM (Fig. 4d,e). A sharp drop in NRM intensity below 100°C in some samples is suggestive of the existence of goethite (Fig. 4b–d). Two polarities were present (Fig. 4 and Fig. 5). In fact, both normal and reversed polarity samples were found to coexist in four sites. The ChRM directions are very close to the present geomagnetic field direction at Guiyang (358.1°/39.2°) or its antipode before tilt correction. This can be more clearly seen at the sample level with a mean in situ direction of D/I =

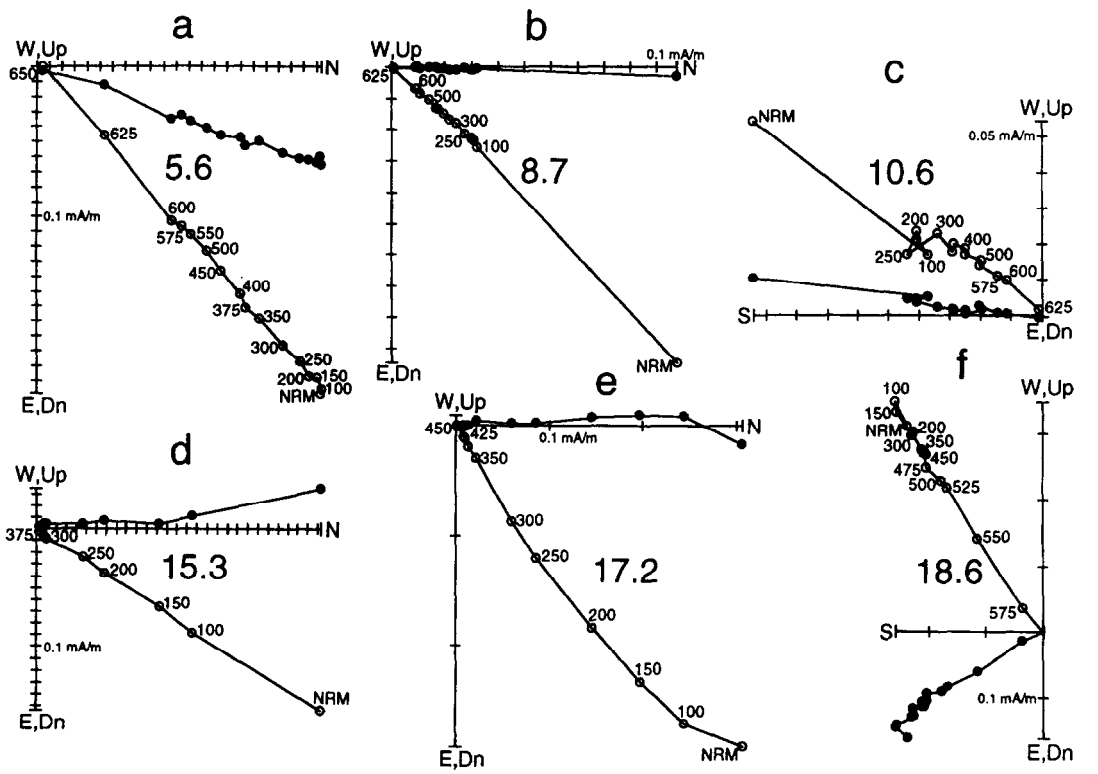


Fig. 4. Representative orthogonal plots of thermal demagnetization for the Huaxi Fm. Coordinates and symbols as in Fig. 2.

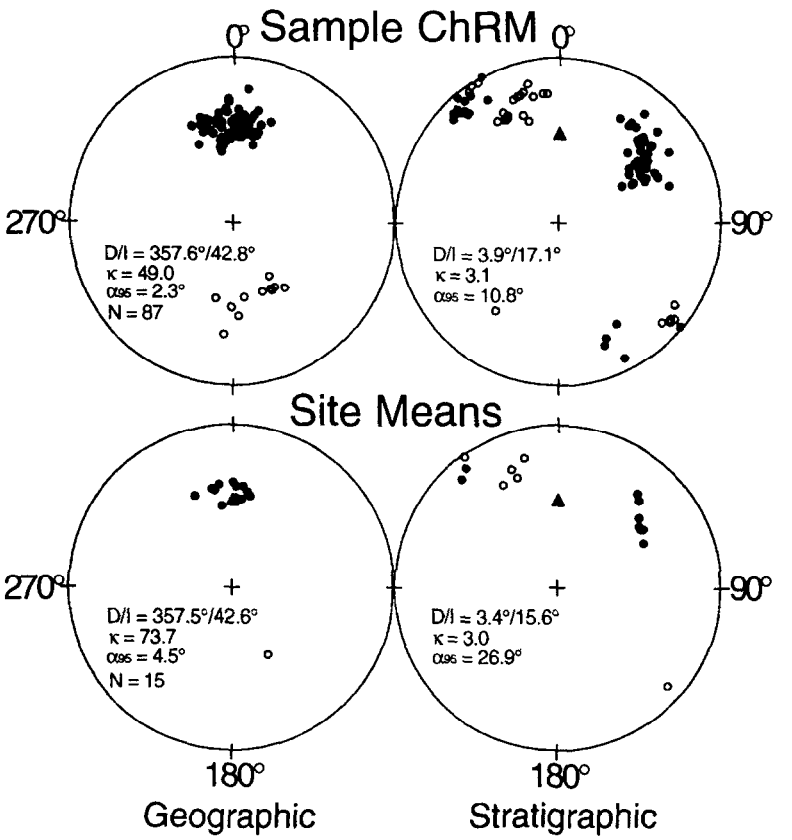


Fig. 5. Equal-area projections of ChRM directions for the Huaxi Fm. Symbols as in Fig. 3.

357.6°/42.8° (Fig. 5, upper). Application of tilt correction split the ChRM directions into three groups dictated by bedding attitudes. The negative fold test both on sample ($\kappa_s/\kappa_g = 0.06$ for N = 87) and site ($\kappa_s/\kappa_g = 0.04$ for N = 15) levels further endorses the recent origin for the ChRM directions resolved from the Huaxi Formation (Fig. 5). The global in situ mean direction based on the 15 sites is D/I = 357.5°/42.6° with $\alpha_{95} = 4.5°$, which corresponds to a pole at 87.1°N/270.2°E with $A_{95} = 5.2°$. The site and formation means computed with Fisherian statistics [9] are given in Table 1.

3.3. The Anshun Formation

As described earlier, the red coloration of the thin bedded dolostone, argillaceous dolostone, brecciated dolostone and salt-solution breccia in the upper part of the Anshun Formation is rather striking among the overwhelmingly gray or light gray carbonates in the studied area, and these rocks were presumably hematite-bearing and magnetically stronger than their light colored counterparts and would thus more possibly hold early magnetizations. For this reason, these rocks were also targeted in 1995 sampling season. Samples were collected at three localities. At the first locality, near a village called Jinguan on the highway from Guiyang to Huangguoshu (Locality 5 in Fig. 1), three sites were drilled from the roadcut, where the rock beds dip to 87–100° east at 35–43°. The second locality is also on the same highway near a village called Haozhitang about 16 km west of downtown Guiyang (Locality 6 in Fig. 1), where three more sites were drilled and the rocks dip toward 49–88° east at 25–27°. The third locality is about 2 km north of downtown Qingzhen (Locality 7 in Fig. 1), four sites were drilled in a quarry, where the bedding dips almost due south at 20–25°. A total

Table 1
Site and formation statistics for Huaxi Formation (26.6°N, 106.7°E)

Site	Bedding	n/N	R/N	In situ		Tilt corrected				Temperature
				D (°)	I (°)	D (°)	I (°)	α_{95}	κ_s	(°C)
1	28/60	6/8	0/6	2.6	38.5	55.8	37.0	12.7	28.6	100–625
3	23/48	5/6	0/5	9.0	40.5	49.0	34.8	8.0	92.8	100–650
4	31/54	7/7	0/5	6.5	44.9	62.7	40.7	3.8	258.5	100–625
5	27/46	5/6	0/7	11.3	42.5	53.0	37.8	5.3	212.9	100–650
8	8/52	6/7	0/6	2.6	44.0	43.1	28.7	7.8	74.5	100–625
9	23/52	5/6	0/5	3.4	44.9	54.6	38.4	7.2	113.0	100–625
11	11/52	3/6	0/3	6.3	38.0	40.5	25.4	11.9	107.7	100–300
12	177/61	6/6	0/6	349.7	40.0	318.3	13.1	2.7	611.8	100–400
13	179/67	5/6	0/5	352.9	36.7	322.8	8.9	6.0	164.7	100–400
14	203/103	5/6	2/3	2.6	45.6	338.7	−23.5	5.7	179.8	100–600
15	204/91	6/6	2/4	0.8	36.2	345.8	−19.1	8.2	67.0	100–600
16	201/98	4/6	0/4	348.8	39.1	340.2	−29.8	8.8	109.7	100–375
17	200/10	66/6	0/6	352.8	49.0	332.2	−29.7	1.9	1271.4	100–450
18	193/50	6/7	5/1	151.0	−51.4	131.1	−10.5	3.5	370.8	100–600
19	200/54	5/6	1/4	337.9	39.8	324.8	−2.3	8.2	88.0	100–575
Formation mean	15/20	1/14	357.5	42.6			3.4	15.6		
				κ_g = 73.4		κ_s = 3.0		α_{95} = 4.5 [a]		

Bedding = strike/dip with dip direction 90° clockwise from strike. n/N = number of samples or sites used/number of samples or sites measured. R/N = number of reverse/normal polarity samples or sites. D, I = declination and inclination. α_{95} = radius of 95% confidence circle about the mean direction. κ_g, κ_s = Fisher precision parameters before and after tilt correction. Temperature = temperature range in which ChRM direction is calculated. [a] For mean in situ direction.

of 61 samples from 10 sites were obtained from the Anshun Formation.

The initial NRM intensities of these samples were indeed stronger than those from the Longtou and Huaxi Formations with the majority in the range 0.2–2.6 mA/m. All the samples from this formation were subjected to stepwise thermal demagnetization, and most of them yielded well defined ChRM directions. Once again, most of the samples displayed univectorial decay behavior during demagnetization (Fig. 6a,b,d); only in a few cases did the samples possess a prominent low temperature component which was directed opposite to the ChRM direction and was removed below 200°C (Fig. 6c). This low temperature component appears to be carried by goethite. The ChRM, on the other hand, is carried by hematite because it persisted to 600°C and higher temperatures in all cases. Two polarities were present with three sites being composed exclusively of reversed samples and four sites containing both normal and reversed samples. The in situ inclinations in some sites are a bit shallower than the present dipole field direction at Guiyang, but the in situ declinations are nearly due north–south in all sites. As a result, the site mean directions fall close to the local present geomagnetic field direction in in situ coordinates, an indication of late remagnetization. This is further supported by the negative fold test both on sample ($\kappa_s/\kappa_g = 0.27$) and site ($\kappa_s/\kappa_g = 0.14$) levels (Fig. 7). The mean in situ direction of the ChRM for the Anshun Formation is D/I = 5.6°/36.7° with α_{95} = 5.1° and N = 10 sites, which yields a paleopole at 82.1°N/242.8°E with A_{95} = 4.6°. Site and formation statistics are given in Table 2.

3.4. Rock magnetic results

To further aid in identification of the magnetic minerals, rock magnetic experiments have been conducted on 27 representative samples from the three formations. They were first stepwise demagnetized in alternating field (AF) up to 100 mT, and then given an anhysteretic remanent magnetization (ARM) in an AF peak field of 100 mT and a biasing field of 55 μT. The ARM was then AF demagnetized in a stepwise fashion. Finally, these samples were given

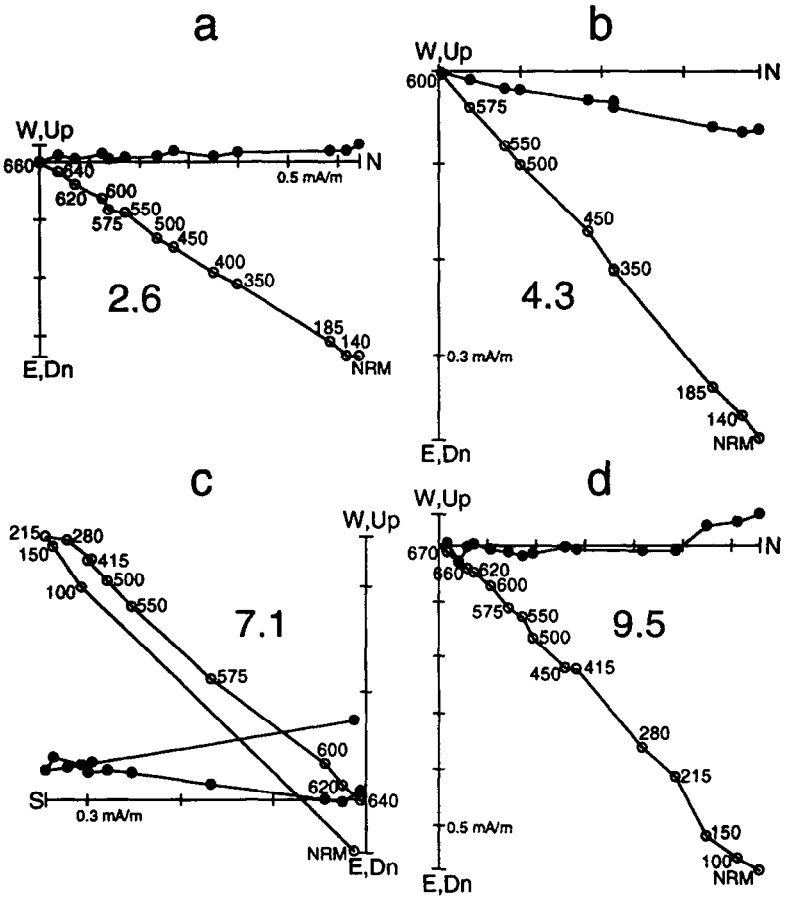


Fig. 6. Representative orthogonal plots of thermal demagnetization for the Anshun Fm. Coordinates and symbols as in Fig. 2.

an incremental isothermal remanent magnetization (IRM) using an impulse magnetizer and were then progressively AF demagnetized.

AF demagnetization had limited effect on the remanence of the samples from the Anshun Formation and those from the Huaxi Formation collected from the east limb of the anticline (Fig. 8c), but very effectively removed the remanence of the samples from the Huaxi Formation collected from most of the west limb sites (Fig. 8b). The response of the samples from the Longtou Formation to AF demagnetization was between the above two extremes (Fig. 8a). Comparison with the thermal demagnetization behaviors of these samples suggests that hematite is the major remanence carrier in the first group while magnetite is the principal magnetic mineral in the second group. The magnetic minerals in the third group are probably a mixture of both. These conclusions appear to be confirmed by the IRM acquisition experiment. The IRM of the first group samples gradually increased with increasing magnetizing field and was not saturated even up to 1.3 T (Fig. 9b) while the second group samples (H12.5, 13.1, 14.4 and 17.2 in Fig. 9a) were saturated at about 200 mT. For the third group (L2.6, 17.5 and 22.5 in Fig. 9a), the increase in IRM intensity was very rapid below 150 mT and then clearly slowed down but was not saturated up to 1.3 T.

Fig. 10 shows the results of the Cisowski [10] test and the modified Lowrie–Fuller test [11] for the magnetite-bearing dolostone and limestone specimens from the Huaxi Formation. The IRM acquisition and demagnetization curves are symmetric and cross over at near 50% of the normalized moment,

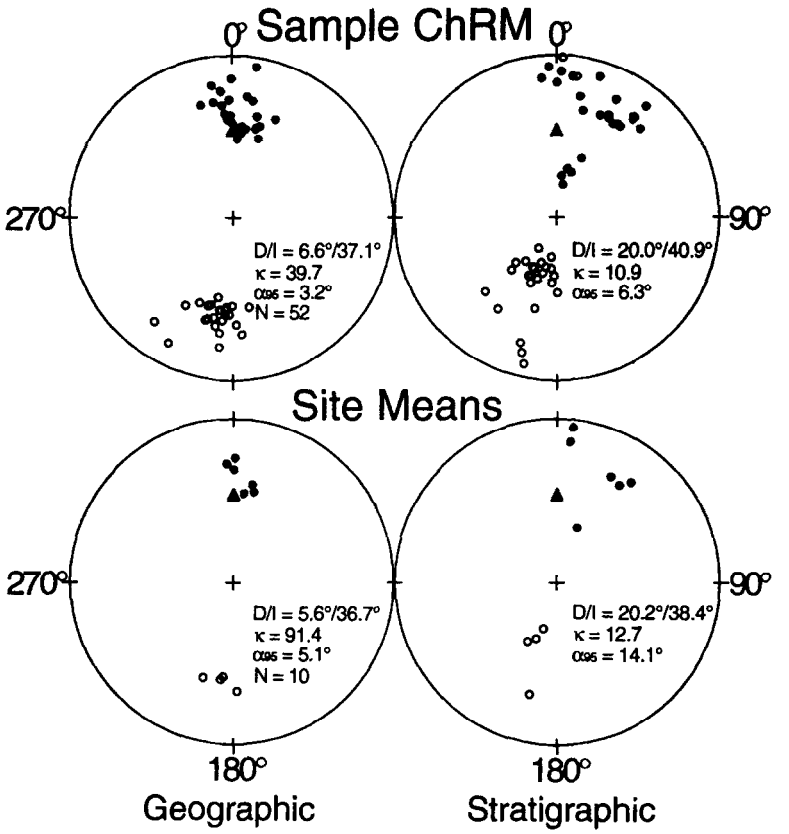


Fig. 7. Equal-area projections of ChRM directions for the Anshun Fm. Symbols as in Fig. 3.

which suggests that the remanence predominantly resides in non-interacting fine (SD or PSD) particles according to Cisowski [10]. On the other hand, ARM is much less resistant to AF demagnetization than IRM, supposedly indicating that the dominant remanence carrier is coarse grained (MD) magnetite [11].

Table 2
Site and formation statistics for Anshun Formation (26.6°N, 106.5°E)

Site	Bedding	n/N	R/N	In situ		Tilt corrected				Temperature
				D (°)	I (°)	D (°)	I (°)	α_{95}	κ_s	(°C)
1	319/27	5/6	2/3	1.1	25.4	6.6	6.1	16.0	23.9	140–660
2	330/25	6/6	0/6	357.3	28.3	5.9	15.0	7.8	74.0	140–660
3	358/26	3/7	2/1	177.5	−33.8	193.9	−30.3	20.5	37.3	140–685
4	87/23	7/7	2/5	11.9	39.4	21.3	61.2	10.2	36.0	140–620
5	91/20	6/6	6/0	197.5	−38.1	205.2	−56.9	11.6	34.4	140–640
6	76/23	6/6	6/0	187.0	−39.1	199.4	−59.7	8.1	69.2	100–620
7	85/25	6/6	6/0	186.0	−41.4	195.3	−65.6	5.1	172.0	150–640
8	10/43	3/5	0/3	0.6	32.0	27.3	28.7	11.9	107.9	150–660
9	358/37	6/7	0/6	13.3	43.1	36.7	25.4	6.8	98.4	150–670
10	357/35	4/6	0/4	7.1	44.8	32.9	30.4	6.6	193.8	100–670
Formation Mean	10/10	4/6	5.6	36.7			20.2	38.4		
				$\kappa_g = 91.4$		$\kappa_s = 12.7$		$\alpha_{95} = 5.1$ [a]		

Explanations as in Table 1.

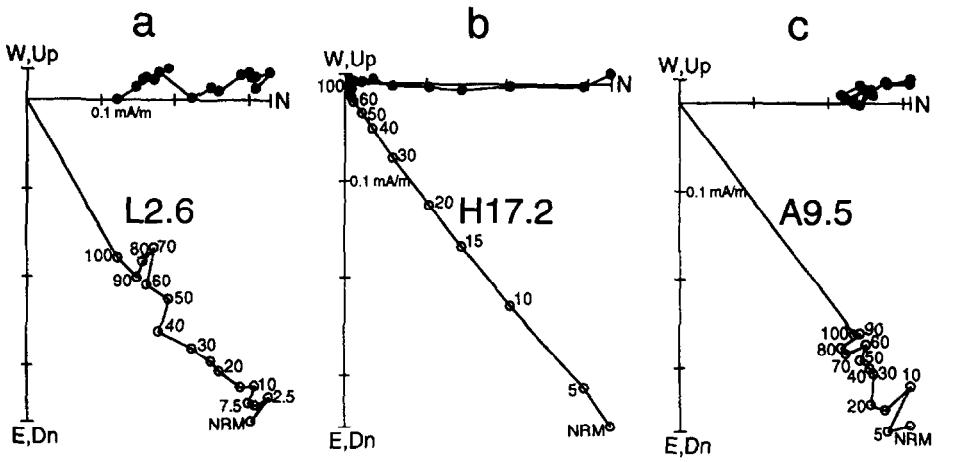


Fig. 8. Orthogonal plots of AF demagnetization for the representative samples studied. L2.6, H17.2 and A9.5 are from the Longtou, Huaxi, and Anshun Formations, respectively.

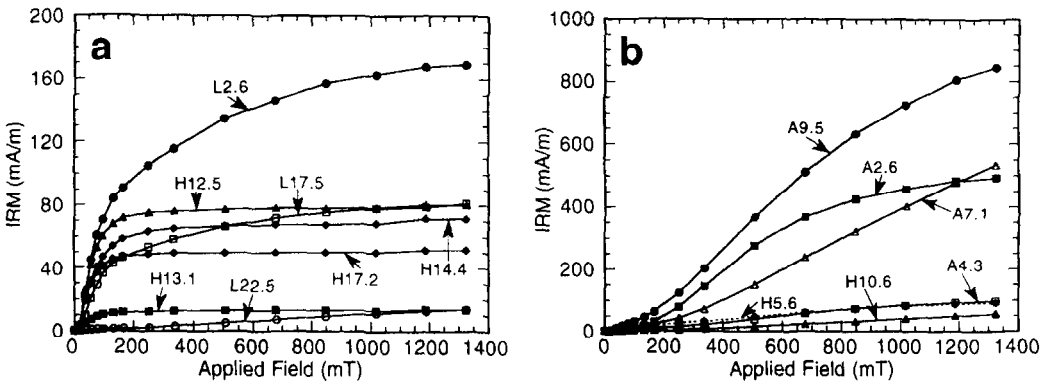


Fig. 9. IRM acquisition curves for representative samples studied. Prefixes L, H and A denote the samples from the Longtou, Huaxi and Anshun Formations, respectively.

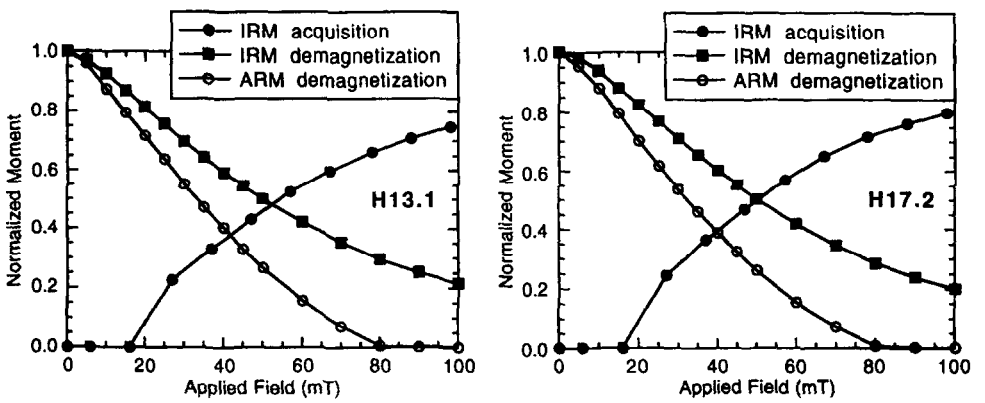


Fig. 10. Cisowski [10] and modified Lowrie–Fuller [11] tests on the magnetite-bearing samples from the Huaxi Fm.

This seems to be in contradiction. Xu and Dunlop [12] have recently shown that the Lowrie–Fuller test does not distinguish domain structure, but it does depend indirectly on grain size. Nevertheless, this kind of behavior has been observed in the remagnetized Paleozoic carbonates from North America [13] and England [14], but is different from that of unremagnetized Mesozoic Tethyan limestones in which ARM and IRM exhibit comparable stabilities against AF demagnetization [15].

4. Discussion

It would be pleasant to report that primary magnetizations have been resolved from the rock formations studied and that the paleomagnetic pole reported by Chan et al. [1] has been reproduced and refined. Unfortunately, this is not the case. Instead, all the ChRMs resolved from the three formations are believed to be remagnetizations of late origin. The strong evidences for this are the resemblance of the in situ directions of the ChRMs to the present geomagnetic or geocentric axial dipole field of the sampling area and the negative fold tests in all cases, despite the presence of dual polarity in the Huaxi and Anshun formations. Incremental unfolding confirms that the ChRMs resolved from the Huaxi and Anshun formations were acquired after the early Cretaceous folding with κ values reaching maxima at 0% unfolding. The proximity of the paleomagnetic poles derived from the Longtou and Huaxi formations to the present geographic North pole and the Miocene pole for the Eurasia suggests that the ChRMs from these two formations were probably acquired in very recent times; the overlap between the paleomagnetic pole from the Anshun Formation and the late Cretaceous poles for the YB indicates that the ChRM from this formation might have been acquired during the late Cretaceous, which is consistent with the shallower mean inclination of the magnetization (Fig. 11). The samples of the Huaxi Fm were taken from a very limited space of 1 m^2 or less at most of the sampling sites. Those of reversed ChRM direction often possessed a normal component which was directed in the present Earth's field, removed below 200°C or 300°C, and possibly carried

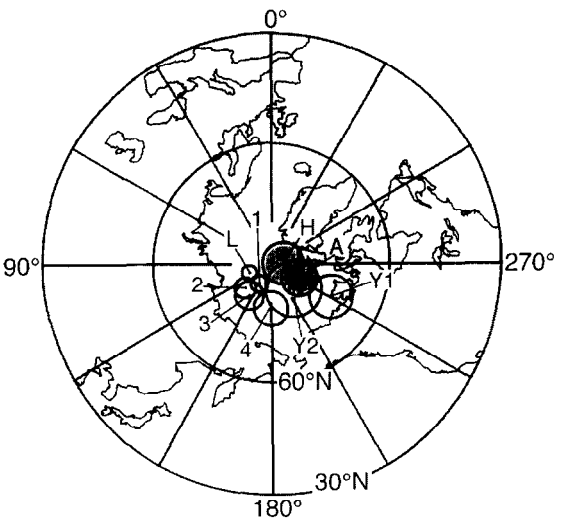


Fig. 11. Comparison of the paleomagnetic poles computed from the remagnetizations of this study (shaded) and the Late Cretaceous poles for the YB and Cenozoic poles for Eurasia. L = pole from the Longtou Fm; H = pole from the Huaxi Fm; A = pole from the Anshun Fm. Late Cretaceous poles for the YB: Y1 from the Sichuan Basin [28]; Y2 from southwestern Sichuan [29]. Cenozoic poles for Eurasia [30]: *1* = 7.9 Ma; *2* = 27.9 Ma; *3* = 42.8 Ma; *4* = 61.3 Ma. Equal-area projection.

by goethite. We regard the presence of both polarities in these sites as an indication of remagnetization.

Kent et al. [16] investigated the paleomagnetism of the Paleozoic to lower Triassic sedimentary rocks from Yichang and Nanjing, most of which are platform carbonates, and found severe late Mesozoic to recent remagnetizations. Zhao [17] documented similar cases in the sedimentary rocks from Hubei, Jiangsu, Zhejiang and Anhui provinces of South China. It appears that the late Mesozoic to Recent remagnetizations are fairly common in South China, especially in the vulnerable platform carbonates. A reinspection of the orthogonal projection of thermal demagnetization of one sample and the stereographic projections of the 'stable' directions from the Huaxi Formation presented by Chan et al. [1] reveals similarities to the results we described earlier: higher unblocking temperature spectra characteristic of hematite and the presence of two polarities. However, five out of eight samples in their study apparently fall in the vicinity of the present Earth's magnetic field or its antipode in in situ coordinates, one of the observations we have made from our collection. Therefore, we suspect that the positive fold test

they claimed is an artifact due to insufficient sampling. This is one of the problems sometimes associated with the preliminary studies.

Widespread late Paleozoic remagnetizations of the early Paleozoic rocks in cratonic North America have been well documented and intensively investigated [18,19]. An explanation for such event by severe weathering when North America was located in tropical latitudes was advocated by Creer [20]. A second hypothesis of thermoviscous remagnetization due to cooling from elevated burial temperatures [21] is considered a mechanism viable only in the areas where thermal maturity is high and maximum burial is likely to have occurred [19]. A third hypothesis of 'orogenic fluids' driven through a large mass of rock as a result of continental collision [22] is very popular and frequently invoked. However, direct link between such fluids and the remagnetization event has not been made [23] and the viability and driving force for regional fluid migration on such a large scale are still questionable [24]. As for the carbonate rocks investigated in this study, the cause(s) for the remagnetization is difficult to ascertain without further petrological and rock magnetic analyses. Weathering is probably the dominant factor responsible for the Cenozoic overprint of the Longtou and Huaxi formations because the age of the remagnetization has been demonstrated to be very recent and the major magnetic carrier is hematite with a possible goethitic precursor. The origin of magnetite in these two formations is unclear, and it may be a product of dolomitization, such as that observed by Hart and Fuller [25] in a Cenozoic dolomite bed, but this would require that the dolomitization take place in very recent times. The remagnetization of the salt-solution breccia and associated rocks in the Anshun Formation must have involved some kind of fluid, but we cannot associate the fluid with orogeny with certainty.

Finally, Wang et al. [26] reported a 'folding-associated remagnetization' from the Middle Triassic platform carbonates and sandstones from Zhengfeng county of southern Guizhou province. The mean direction of the remagnetization they reported is nearly north–south with very shallow inclination and mixed polarities. The age for the remagnetization they inferred is Late Triassic/Jurassic. Based on this and similar paleomagnetic directions they observed from Nanjing [27], they proposed a new loop to be added to the APWP for the YB. Zhengfeng is 180 km southwest of Guiyang and is situated in the same Triassic (Central Guizhou) stratigraphic subprovince as Guiyang [6]. However, such 'widespread' remagnetization has not been observed in our sample collection. The folding age they cited for the Triassic rocks they studied is poorly constrained, as they noted, and may well be the early Cretaceous, as in Guiyang. If this is true, the remagnetization direction they reported would be at odds with those we observed from the Anshun Formation from Guiyang and Kent et al. [15] observed from the Paleozoic rocks from Yichang and Nanjing. One possible cause for this discrepancy is that the area Wang et al. sampled is not part of the YB but part of the Huanan nappe, as they also noted. However, the number of usable samples from their collection is relatively small; therefore, the results they reported need to be verified before this discrepancy can be resolved.

5. Conclusions

Severe late Cretaceous to Recent remagnetizations have been revealed from the Middle and Lower Triassic platform carbonates from the vicinity of Guiyang and thus discredit a Middle Triassic paleomagnetic pole reported by Chan et al. [1] from the same region. One of the problems associated with some preliminary paleomagnetic investigations is insufficient sampling; therefore, caution should be exercised when evaluating and using the preliminary results, especially those derived from platform carbonates which are usually vulnerable to late remagnetization.

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